

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<tool_call> reasoning process here

The de Broglie wavelength λ of a particle is given by :

$$\lambda = h / p$$

where :

- h is Planck's constant = 6.626×10^{-34} J $\cdot$ s (or 6.626×10^{-27} erg $\cdot$ s),
- p is the momentum of the particle .

For a non-relativistic electron , the kinetic energy K is related to momentum by :

$$K = p^2 / (2m)$$

where m is the electron mass = 9.109×10^{-31} kg .

We are given K = 1 eV = 1.602×10^{-19} J .

Step 1 : Find momentum p :

$$p = \sqrt{2mK}$$

$$\begin{aligned} p &= \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}} \\ p &= \sqrt{2.918 \times 10^{-50}} \\ p &\approx 5.402 \times 10^{-25} \text{ kg}\cdot\text{m/s} \end{aligned}$$

Step 2 : Compute de Broglie wavelength :

$$\begin{aligned} \lambda &= h / p = (6.626 \times 10^{-34}) / (5.402 \times 10^{-25}) \\ \lambda &\approx 1.227 \times 10^{-9} \text{ m} \end{aligned}$$

Convert to cm : 1 m = 100 cm , so :

$$\lambda \approx 1.227 \times 10^{-7} \text{ cm}$$

Now check the options for value near 1.227×10^{-7} cm .

Option G : 1.23×10^{-7} cm $\approx$ matches closely .

Thus , the correct answer is G .

<answer>G</answer>